

AF2122_97 OK

Spoligotype	SB0140
Octal	664073777777600
Hexadecimal	6D-03-5F-7F-FF-60
Binary	110110100000111011111111111111111111100000
Pattern	
Data	reads (fastq, single-end)
Input file	/example/data/AF2122_97.fastq.gz 200.2 MB · modified 2026-09-24 13:13:56 UTC · MD5 f4280874bc6b93749d8e7f10935bce27
Input size	1,212,727 reads, 284,264,369 bases
Estimated depth	65x (all bases / 4.4 Mb genome)
Minimum count	5 reads per spacer to call it present
Present spacers	30 of 43, median count 50
Run time	7.5 s

Species and lineage

Species	M. bovis
Lineage	BOV · M. bovis · typical spoligotypes: BOV_2, BOV_1
MTBC DNA	median 80.5 reads per control region, 100% of the control regions found; about 100% of the reads are MTBC

Region	Deleted in	Depth relative to MTBC control	Call
RD9	M. africanum and animal lineages (incl. M. bovis)	0.00	deleted
RD4	M. bovis and BCG	0.00	deleted
RD1	BCG (and M. microti)	0.80	present

Lineage SNP	Position (H37Rv)	Gene	Reads with lineage allele	Other reads
BOV_AFRI	1,882,180	Rv1662	62	0
BOV	2,831,482	Rv2515c	56	0

Reads per spacer

Spacer	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Count	56	47	0	58	63	0	76	0	0	0	0	0	50	43	38	0	36	29	45	50	47	51
Spacer	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	
Count	50	54	140	74	73	78	57	56	50	46	39	41	45	45	54	53	0	0	0	0	0	

Blue: present (count ≥ 5). Orange: called absent but seen in some reads.

NC_002945 OK

Spoligotype	SB0140
Octal	664073777777600
Hexadecimal	6D-03-5F-7F-FF-60
Binary	110110100000111011111111111111111111100000
Pattern	
Data	assembly (fasta)
Input file	/example/data/NC_002945.fasta 4.2 MB · modified 2026-09-24 13:13:56 UTC · MD5 f07a5427fbfa08b83cf37a8e346250c
Input size	1 contig, 4,349,904 bases
Minimum count	1 contig per spacer to call it present
Present spacers	30 of 43, median count 1
Run time	2.0 s

Species and lineage

Species	M. bovis
Lineage	BOV · M. bovis · typical spoligotypes: BOV_2, BOV_1
MTBC DNA	median 1 contigs per control region, 100% of the control regions found

Region	Deleted in	Depth relative to MTBC control	Call
RD9	M. africanum and animal lineages (incl. M. bovis)	0.00	deleted
RD4	M. bovis and BCG	0.00	deleted
RD1	BCG (and M. microti)	1.00	present

Lineage SNP	Position (H37Rv)	Gene	Reads with lineage allele	Other reads
BOV_AFRI	1,882,180	Rv1662	1	0
BOV	2,831,482	Rv2515c	1	0

Contigs per spacer

Spacer	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Count	1	1	0	1	1	0	1	0	0	0	0	0	1	1	1	0	1	1	1	1	1	1
Spacer	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	
Count	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	

Blue: present (count ≥ 1). Orange: called absent but seen in some contigs.

Species and lineage

Region	Deleted in	Depth relative to MTBC control	Call
RD9	M. africanum and animal lineages (incl. M. bovis)	1.00	present
RD4	M. bovis and BCG	1.00	present
RD1	BCG (and M. microti)	1.00	present

Contigs per spacer

Blue: present (count ≥ 1). Orange: called absent but seen in some contigs.

Run information

Operator	Example
User	example
Computer	example
Operating system	Linux-6.8.0-139-generic-x86_64-with-glibc2.39
Started	2026-09-24 13:33:07 UTC
Finished	2026-09-24 13:33:19 UTC
Duration	11.8 s
Working directory	/example
Command	spoligotyper -i data/ -o results/ -t 8 --operator Example

Parameters

Input	/example/data
Output folder	/example/results
Minimum count	default (5 for fastq, 1 for fasta)
Threads	8
Jobs	1
Seal memory	1g
Species and lineage	yes

Software

spoligotyper	0.4.2
Python	3.14.7 (/opt/conda/bin/python3.14)
Seal	/opt/conda/bin/seal.sh
BBTools	40.02
Java	openjdk version "25.0.2-internal" 2026-01-20

Reference data

Spoligotype database	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/spoligotype_db.txt 1,976 patterns · MD5 0a88f7b89f746de2a11e1b4af0151e36
Spacer sequences	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/spoligo_spacers.fasta 43 spacers · MD5 033c08e66aaa90193973f8d542e94c1
Species markers	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/markers.fasta MD5 55e4a9ef7d10c9a241b4a8ac4eb55200
Lineage SNP barcode	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/lineage_barcode.tsv MD5 28821eba812f42d2c0e9ce81053ae6cf
Lineage SNP sequences	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/lineage_snps.fasta MD5 7e662a9baee450e79065705eb7b79021

Method

Spacers were counted with Seal (BBTools): each of the 43 spacers is searched as a single 25-mer on both strands, allowing 1 mismatch ($k=25$, $hdist=1$, $rcomp=t$, $maskmiddle=f$, $ambiguous=all$). A spacer is present when it is found in at least the minimum count of reads (contigs for assemblies). The binary pattern is converted to the octal code (Dale et al. 2001) and the hexadecimal code, and looked up in the spoligotype database for its SB number. Species: the read depth of regions of difference RD9, RD4 and RD1 (100 bp segments, same Seal parameters) is compared with the depth of MTBC-specific control regions; a region is deleted when its relative depth is at most 0.1, present when it is at least 0.5. The fraction of MTBC reads is the control depth divided by the depth expected from the number of bases. Lineage: reads carrying each allele of the 62 SNPs of the Coll et al. (2014) barcode are counted with exact 31-mers ($k=31$, $hdist=0$); a lineage is called when at least 80% of the reads (and at least 3, or 1 contig) carry its allele.

References

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Dale JW et al. Spacer oligonucleotide typing of bacteria of the Mycobacterium tuberculosis complex: recommendations for standardised nomenclature. Int J Tuberc Lung Dis 5:216-219 (2001).

Smith NH, Upton P. Naming spoligotype patterns for the RD9-deleted lineage of the *Mycobacterium tuberculosis* complex; www.Mbovis.org. *Infect Genet Evol* 12:873-876 (2012). doi:10.1016/j.meegid.2011.08.002

Brosch R et al. A new evolutionary scenario for the *Mycobacterium tuberculosis* complex. *Proc Natl Acad Sci USA* 99:3684-3689 (2002). doi:10.1073/pnas.052548299

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Duceppe M-O. spoligotyper 0.4.2: in silico spoligotyping of *Mycobacterium tuberculosis* complex genomes. Zenodo. <https://doi.org/10.5281/zenodo.22926160>